Paleomagnetism of the Beaver Bay Complex, Minnesota

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The Beaver Bay complex is a series of intrusions of gabbroic rocks related to the Middle Keweenawan Duluth gabbro complex. The Beaver Bay ferrogabbro, one of three units into which the complex has been divided, is an iron-rich gabbro with a very prominent igneous lamination. Magnetic anisotropy caused by preferential alignment of inequant magnetite grains within the lamination appears to have deflected the direction of remanent magnetization of the ferrogabbro by about 8.5°. When samples from the ferrogabbro are eliminated, the resulting paleomagnetic pole for the Beaver Bay complex falls at 27.5°N, 170.5°W, where it forms part of a distinct cluster of paleomagnetic poles for Keweenawan igneous rock units of negative (normal) polarity. Positively magnetized Keweenawan igneous rocks, which appear to be older than the negatively magnetized units, give paleomagnetic pole positions that are scattered over a considerable area to the northeast of the cluster of negative poles. Magnetic intensities vary markedly throughout the complex, probably in response to the ordinary processes of magmatic differentiation.

INTRODUCTION

The Beaver Bay complex of *Grout and Schwartz* [1939] was named for exposures of gabbroic rocks, with granophyre and inclusions of anorthosite, found along the shore of Lake Superior in southern Lake County, Minnesota (Figure 1). Most observers correlate these rocks with the closely underlying Duluth gabbro complex (middle or late Keweenawan). A potassium-argon age of 1000 m.y. [*Goldich et al.*, 1961, p. 53, sample KA 92B] supports this correlation. The Beaver Bay complex intrudes the upper part of the middle or upper Keweenawan lava sequence as a series of gently dipping, roughly concordant sheets, with a few crosscutting bodies. The general geology and the tectonic setting of the Keweenawan rocks in the Lake Superior area have been described recently by *White* [1966] and *Halls* [1966].

Interest in the Beaver Bay complex was stimulated by the discovery [*Grout and Schwartz*, 1939, p. 35, Table 2, analysis 2] of iron-rich gabbro similar to that found in the Skaergaard intrusion [*Wager and Deer*, 1939]. *Muir* [1954] made a detailed study of the crystallization history of ferromagnesian minerals from these iron-rich rocks and observed iron-enrichment trends approximating those of the Skaergaard intrusion and the iron-rich dolerite of the New Amalfi sheet, South Africa [*Poldervaart*, 1944]. Later, *Gehman* [1957] subdivided the Beaver Bay rocks into three distinct units on the basis of significant lithologic differences. Gehman's units and age relations provide the geologic background for this research, and we are indebted to him for permission to use his unpublished material. Gehman's three units are as follows.

Beaver River gabbro: The Beaver River gabbro of *Gehman* [1957] is the oldest and most widespread unit of the Beaver Bay complex, but it shows the least chemical differentiation. The unit, which is massive and of variable grain size, is distinguished in the field by its mottled appearance, caused by prominent ophitic 'patches' formed by the weathering of poikilitic titanaugites. Inclusions of anorthosite are common in the Beaver River gabbro, especially near its contacts, but are found rarely or not at all in the remainder of the Beaver

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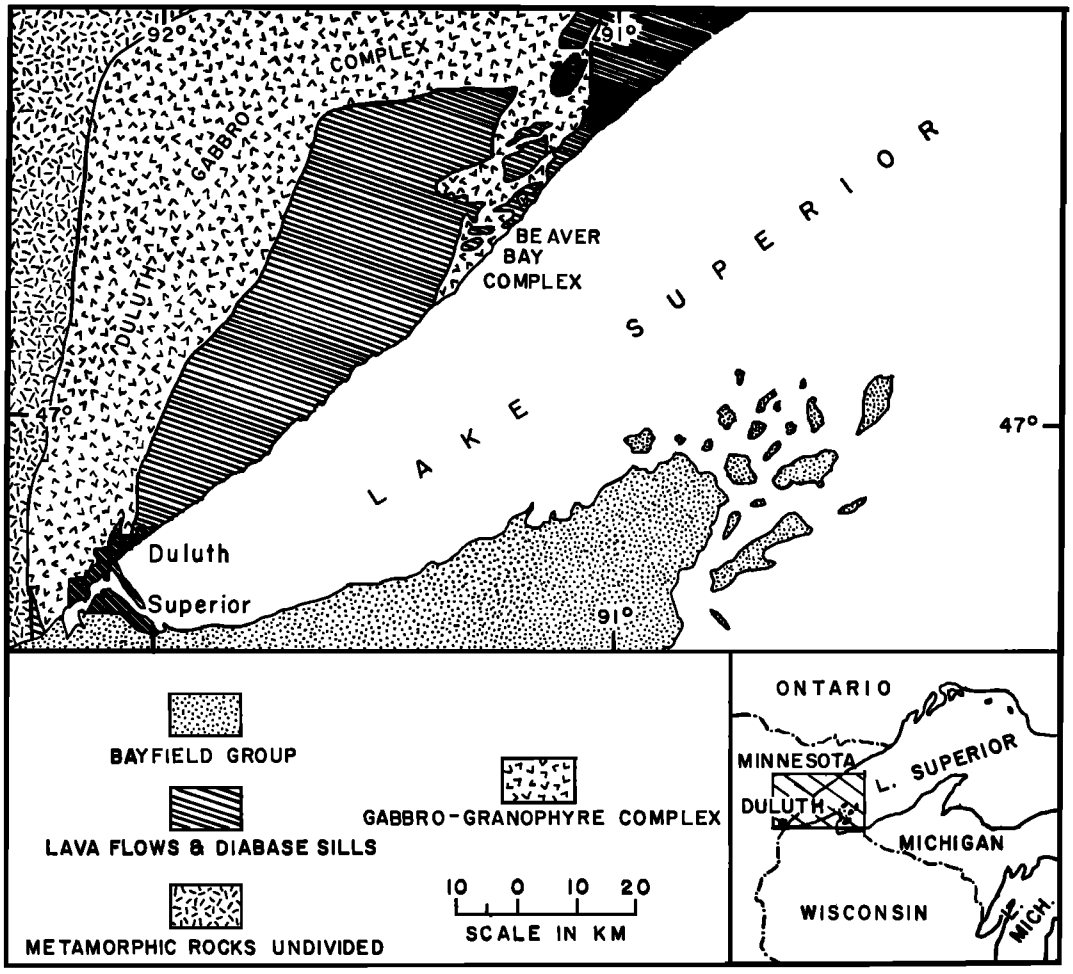


Fig. 1. Index map of the western tip of Lake Superior. Modified from *White* [1966].

Bay complex. The structure of the main part of the Beaver River gabbro is that of a concordant sheet, dipping gently toward Lake Superior. The Beaver River gabbro shows surprisingly little chemical differentiation in view of its size and textural heterogeneity. *Gehman* [1957, p. 81] attributes this to rapid cooling following intrusion into relatively cold country rock.

Beaver Bay ferrogabbro: The Beaver Bay ferrogabbro of *Gehman* [1957] is strongly differentiated, in a manner similar to the more iron-rich portions of the Skaergaard stratiform complex. It outcrops as two separate bodies, each less than 4 km in maximum horizontal dimension. Horizontal igneous lamination is strongly developed, and cryptic layering causes the rock to vary in composition from hortonolite- to fayalite-ferrogabbro. Iron content (total iron calculated as FeO) ranges from 18 to 21.5%, compared with 10 to 11% in the Beaver River gabbro. Despite strong chemical and mineralogical differentiation, the Beaver Bay ferrogabbro shows little variation in texture, grain size, or gross appearance in the field. Rhythmic layering is present in a few exposures. Although the base of neither ferrogabbro body is exposed, the layering and lamination suggest that both are floored intrusions.

Black Bay gabbro: The Black Bay gabbro of *Gehman* [1957] is the youngest and smallest of Gehman's three units and is found chiefly

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in two large dikes surrounding the ferrogabbro intrusions. Its contacts with the ferrogabbro are unchilled, indicating that the Black Bay gabbro must have followed the ferrogabbro closely in time. The Black Bay gabbro is medium- to coarse-grained and is characterized by numerous pegmatite zones and by marked variations in texture. Like the Beaver River gabbro, the Black Bay gabbro shows no igneous lamination or other planar structure. According to *Gehman* [1957, pp. 78–80] the Black Bay gabbro has chemical and mineralogical similarities to the dolerite-pegmatites of *Walker* [1953], which are believed to be late differentiation products of gabbroic magma. Gehman points out, however, that the Black Bay gabbro cannot represent a late differentiate of the ferrogabbro magma, despite gross similarities in mineralogy and differentiation trend, because the Black Bay gabbro contains plagioclase and augite of higher melting point than do the phases of ferrogabbro that it cuts. The total iron content (as FeO) of the Black Bay gabbro is 13.5% [*Gehman,* 1957, p. 72, one analysis].

Our interest in the Beaver Bay complex originally focused on the Beaver Bay ferrogabbro, which seemed to offer an excellent opportunity to test the effect of extreme iron enrichment on certain magnetic properties, such as intensity and stability of remanent magnetization and magnetic susceptibility, which determine in situ rock magnetism. Also, we wished to investigate the effect of the strong horizontal lamination of the ferrogabbro on its average direction of remanent magnetization. For these reasons, a disproportionate number of samples was collected from the ferrogabbro exposures. The Beaver River and Black Bay gabbros also were sampled, both to furnish a 'standard' of more typical gabbroic lithology with which to compare the ferrogabbro and to provide a paleomagnetic pole position for the Beaver Bay complex which would be free of the effects of anisotropy.

Field and Laboratory Procedures

One hundred thirty samples were collected from sixty-one sites in a belt 1 km in width extending approximately 16 km along the NW shore of Lake Superior. Two samples were collected at most sites, although at a few sites it was necessary to take as many as five samples in order to include the full range of lithologic variation. Fresh exposures were available in wavecut terraces and bluffs along the lake front and in road cuts along North Shore drive; farther inland most exposures were extensively weathered and were not sampled. Samples were drilled in place and were oriented by means of a standard geologist's pocket transit. Difficulties in the orientation process arose because the area is underlain by strongly magnetic rock, some portions of which have been struck and intensely remagnetized by lightning. Fortunately, the open location of most of the sites made it possible to correct for declination anomalies by sighting on distant landmarks or cultural features of known bearing. In this way all but about 5% of the specimens probably are oriented to within 3°, whereas the remainder may have errors that probably do not exceed 5°. The need to check for errors in magnetic declination at each outcrop made it easy to avoid areas that had been strongly affected by lightning. Such areas were characterized by large declination anomalies (up to 180°) and strong declination gradients (as high as a few degrees per centimeter).

Direction and intensity of each sample were measured on a motor-driven spinner magnetometer patterned after the instrument described by *Doell and Cox* [1965]. The accuracy of measurements on this instrument is about 2°. To minimize the effects of secondary components of magnetization, all specimens were progressively demagnetized in alternating magnetic fields of 37.5-, 75-, 150-, and 300-oe peak intensity. Samples from five sites in the Beaver Bay ferrogabbro proved to be totally unstable (directions of individual specimens were scattered initially, changed erratically during demagnetization, and failed to approach a common direction). These sites were eliminated from subsequent calculations. Table 1 shows the average direction of magnetization for each lithologic unit at each step in the demagnetization process.

It is evident from Table 1 that alternating-field magnetic cleaning had considerable influence on the dispersion of directions of magnetization in the three gabbro units but has produced very little change in their average directions. In general, dispersion decreased un-

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TABLE 1. Directions of Magnetization in the Beaver Bay Complex

$\bar{H}$	Unit	N	D, deg	I, deg	k	α_{95}	δ, deg
NRM	BL	11	292.5	+48.5	12.0	13.5	22.0
	F	27	291.0	35.5	6.0	12.0	32.0
	BR	18	282.5	48.5	101.0	3.5	8.0
37½	BL	11	284.5	47.0	16.5	11.5	19.0
	F	27	286.0	37.0	28.5	5.5	15.0
	BR	18	280.5	45.0	106.0	3.5	7.5
75	BL	11	288.0	48.0	18.0	11.0	18.0
	F	27	284.5	36.5	82.5	3.0	9.0
	BR	18	279.0	43.5	102.5	3.5	8.0
150	BL	11	287.0	48.0	13.5	13.0	21.0
	F	27	282.0	34.0	39.5	4.5	12.5
	BR	18	277.5	41.5	52.5	5.0	11.0
300	BL	11	280.5	46.0	6.0	20.5	31.5
	F	27	279.0	32.0	12.5	8.0	22.5
	BR	18	272.5	39.0	16.5	9.0	19.5
75	BL and BR combined	29	282.5	45.0	36.0	4.5	13.5

Explanation

$\bar{H}$ is the peak value of the alternating magnetic field used in magnetic cleaning (NRM, natural remanent magnetization).

Unit designations: BL, Black Bay gabbro; F, Beaver Bay ferrogabbro; BR, Beaver River gabbro.

N is the number of site-mean directions of magnetization involved in each calculation. D, I, k, α_{95} are declination, inclination, best estimate of the precision parameter, and 95% confidence limits, calculated by the method of *Fisher* [1953]. δ is the angular standard deviation [*Wilson*, 1959].

til the 75-oe demagnetization level was reached and then began to increase (for possible causes of such an increase, see *Doell and Cox* [1967]). Following procedures suggested by *Irving et al.* [1961], we have selected 75 oe as the demagnetization level that yields the best available record of the original direction of TRM in the rock.

A choice of 75 oe as optimum peak field for magnetic cleaning accords well with an analysis based on individual specimen directions. More than 90% of specimens from stable sites reached a stable end point [*McElhinny and Gough*, 1963] at or below 75 oe. Increasing scatter beyond 75 oe is due primarily to a few specimens that begin to diverge from the average direction in fields of 150 oe or above. These specimens have been demagnetized in peak fields ranging up to 600 oe, and they do not appear to approach a common direction. Elimination of these specimens would not significantly alter the average directions listed in Table 2.

Magnetic Anisotropy

Also obvious from Table 1 is the fact that the average direction of magnetization of the Beaver Bay ferrogabbro is systematically shallower (has a lower inclination) than in the other two units. This is illustrated in Figure 2, which shows equal-area plots of directions of magnetization in the three units, and in Figure 3, an equal-area plot of their average directions of magnetization with associated confidence limits. In these illustrations data from the Beaver River gabbro and the Black Bay gabbro have been combined in order to contrast the massive portion of the Beaver Bay complex with the strongly laminated ferrogabbro phase. Combining the data from the Beaver River and Black Bay gabbros in this way is justified by the fact that the two units are almost identical in age and show no significant difference in their average directions of remanent magnetization (95% probability) at any stage of magnetic cleaning.

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TABLE 2. Paleomagnetic Pole Positions for Igneous Rocks of Probable Keweenawan Age in the Lake Superior Area

Symbol Fig. 5	Unit	Sampling Locality λ, °N	Sampling Locality φ, °W	D, deg	I, deg	k	N	Polarity	Paleomagnetic Pole λ	Paleomagnetic Pole φ	Reference
1.34	Baraga County, Michigan, dikes	46.5	88.5	82.0	−86.0	82	36	+	45.0°N	99.5°W	*Graham* [1953]
1.38	Basalt in Copper Harbor conglomerate	47.0	88.5	289.5	+37.0	16	13	−	28.0°N	178.0°W	Recalculated from *DuBois* [1962]
1.39	Portage Lake lava series	47.0	88.5	282.5	+44.0	47	31	−	26.5°N	168.5°W	*DuBois* [1962]
	Portage Lake lava series	47.0	88.5	286.0	+33.0		14	−	24.0°N	177.0°W	*Vincenz* [1968] *Vincenz and Yaskawa* [1968]
1.79	Duluth gabbro complex	47.0	92.0	285.5	+33.0	17	18	−	23.5°N	179.5°E	*DuBois* [1962]
1.80	Mamainse Point lavas	47.0	84.5	292.0	+40.5	19	12	−	31.5°N	173.5°W	*DuBois* [1962]
1.81	Logan diabase, I	48.5	89.0	117.0	−76 0	13	80	+	53 5°N	130 0°W	*DuBois* [1962]
1 82	Logan diabase, II	48.0	89.5	288.5	+47 5	21	25	−	33 0°N	171.5°W	*DuBois* [1962]
1 84	Alona Bay lavas	47.0	84.5	49.5	−84.5	12	12	+	39.0°N	95.0°W	*DuBois* [1962]
M	Gabbro near Mellen, Wisc.	46.5	91.0	293.5	+47.5	14	14	−	36.0°N	175.5°W	*Books et al.* [1966]
B	Beaver Bay complex	47.0	91.5	282.5	+45.0	36	29	−	27.5°N	170.5°W	This paper

Explanation

Numerical symbols identify poles listed in *Irving* [1964. Appendix]. λ and φ are latitude and longitude, respectively. *D*, *I*, and *k* are explained in Table 1. *N* is number of samples for all but poles M and B, where it is the number of sampling sites. Polarity follows the convention of *Irving* [1964, p. 71].

The Beaver Bay complex probably cooled and acquired its remanent magnetization substantially as a single unit, rather than as a series of distinct intrusions. This is supported by the following considerations: (1) The gabbro was intruded at considerable depth into a part of the earth's crust that was experiencing intense igneous activity. Thus, cooling to the Curie temperature probably was a slow process, and a relatively long period of time should have been required for the entire complex to acquire its remanent magnetization. (2) The successive intrusions that make up the Beaver Bay complex followed one another closely in time. *Gehman* [1957, p. 81] observed that only the Beaver River gabbro has chilled margins, so that presumably the country rock still was hot when the later bodies were intruded. (3) The Curie temperatures of magnetic oxide minerals commonly found in gabbroic rocks are much lower than the solidification temperature of gabbro; consequently, the earliest intrusive pulses of the Beaver Bay complex must have been largely unmagnetized when the latest pulse arrived. Thus, the three rock units that make up the Beaver Bay complex very likely acquired their remanent magnetization during substantially the same period of time and in substantially the same configuration of the geomagnetic field, and the systematic divergence of the ferrogabbro probably indicates that it did not become magnetized parallel to the ambient field. In particular, magnetic anisotropy is suggested by the fact that the ferrogabbro has a strong horizontal lamination and the divergence of its average direction of magnetization consists almost entirely of a deflection toward the horizontal.

It is possible to test the hypothesis that magnetic anisotropy on the part of the ferrogabbro could have caused the observed deflection of its direction of remanent magnetization toward the horizontal. The method used follows that of *Uyeda et al.* [1963]. X-ray diffraction studies indicate that nearly the entire magnetic fraction of the ferrogabbro consists of relatively pure magnetite; therefore, crystalline anisotropy probably can be neglected. To estimate grain-shape anisotropy, thin sections of ferrogabbro were cut approximately normal to the

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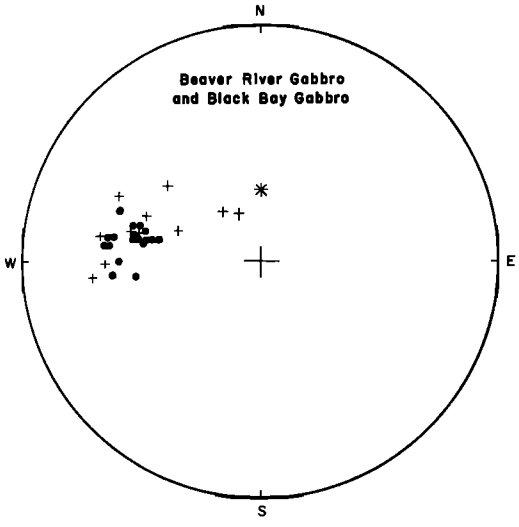


Fig. 2*a*.

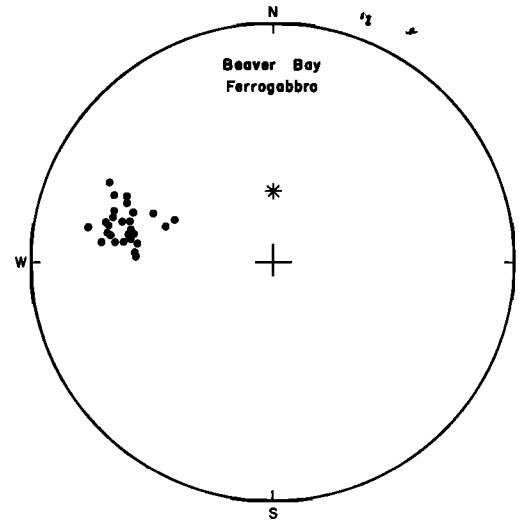


Fig. 2*b*.

Figure 2 shows the site mean directions of magnetization in the Beaver Bay complex after magnetic cleaning in a peak field of 75 oe. Asterisk shows direction of present dipole field. (*a*) Crosses indicate Black Bay gabbro, dots indicate Beaver River gabbro.

plane of lamination, and the ratio of grain diameter parallel and normal to the lamination measured for 136 opaque grains. The mean value of this dimension ratio was 1.38. Using this value plus a measured bulk susceptibility for the magnetic grains in the ferrogabbro of 0.10 emu/cm^3, a theoretical deflection angle can be calculated from equation 11 of *Uyeda et al.* [1963, p. 287]

$$\tan \theta_{obs} = \tan \theta_{ex}/P(T_0)P(T_b)$$

In this equation θ_{obs} is the angle of inclination of the average direction of magnetization of the ferrogabbro, θ_{ex} is the inclination of the paleogeomagnetic field, as recorded by the nonlaminated gabbro bodies, and P is the anisotropy

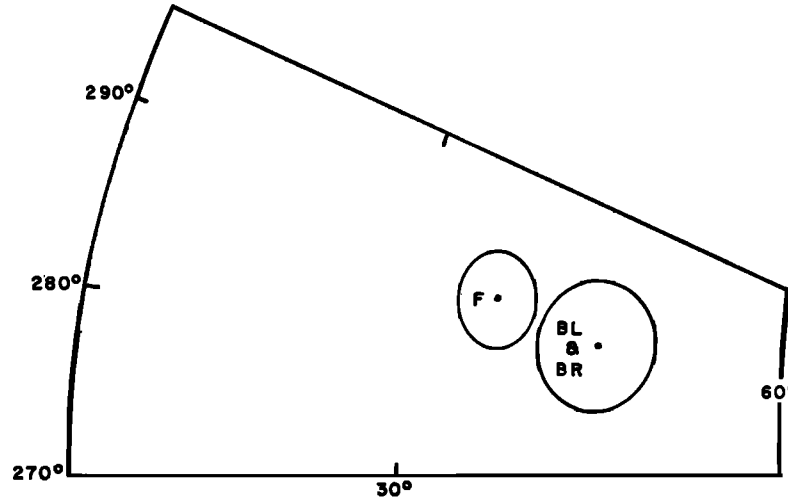


Fig. 3. Average directions of magnetization and ovals of confidence (95% probability level), after magnetic cleaning in a peak field of 75 oe. F, Beaver Bay ferrogabbro; BL, Black Bay gabbro; BR, Beaver River gabbro.

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factor evaluated at surface temperature (T_0) and at the blocking temperature (T_b) of the magnetic grains in the rock. P can be obtained directly from measurements of directional susceptibility or, as in this study, can be estimated from the effective dimension ratio of the magnetic grains.

The angle θ_{obs} (38.8°) calculated by this method compares closely with the actual inclination of the ferrogabbro (36.7°). The calculation required the following assumptions. (1) The shape of the average magnetic grain was assumed to be that of an oblate spheroid ($b/a = 1.38$) with its equatorial plane in the plane of igneous lamination. (2) P, the anisotropy factor, was assumed to be independent of temperature: $P(T_0) = P(T_b)$. Assumption 1 is justified by the observation that no lineation could be observed within the plane of igneous lamination, indicating that in the lamination plane the average grain diameter is nearly independent of direction. Assumption 2, which implies that magnetic susceptibility below the Curie temperature is independent of temperature (Hopkinson effect absent), probably is in error by as much as 20%. A 20% rise in magnetic susceptibility between T_0 and T_b would, however, actually effect a slight improvement in the agreement between the measured and the calculated values. These calculations seem to be sufficiently precise to show that magnetic anisotropy arising from a high degree of preferred orientation of inequant magnetite grains could have produced a deflection of about 8.5° in the average direction of remanent magnetization of the ferrogabbro.

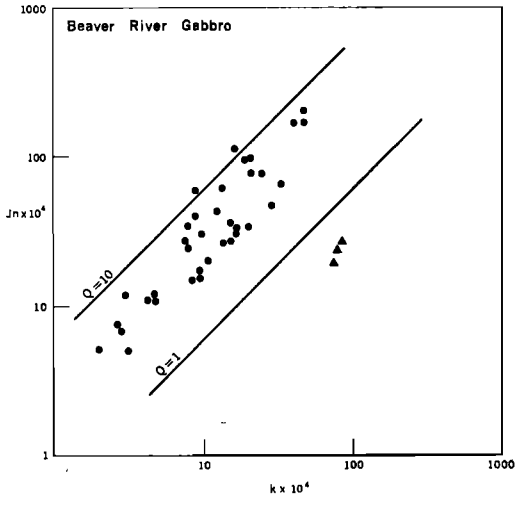


Fig. 4*a*.

Intensity of Magnetization

Magnetic intensities of samples from the three gabbro units are shown in Figure 4 in the form of a series of plots of natural remanent mag-

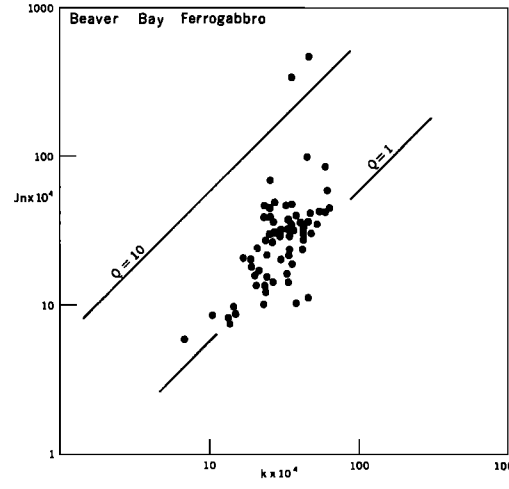


Fig. 4*b*.

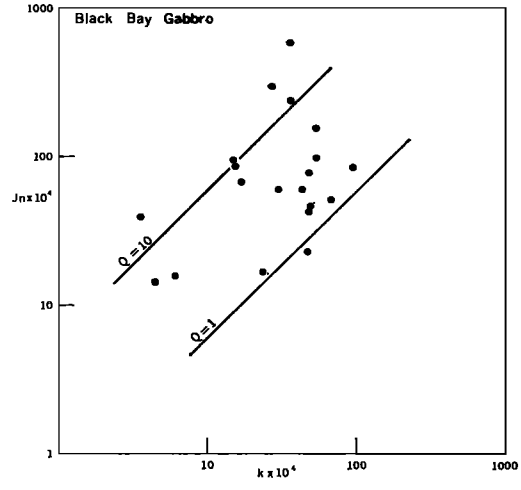


Fig. 4*c*.

Figure 4 shows the plots of intensity of natural remanent magnetization (J_n) versus magnetic susceptibility (k) for (*a*) the Beaver River gabbro, (*b*) the Beaver Bay ferrogabbro, and (*c*) the Black Bay gabbro. Q is the ratio J_n/kT [*Koenigsberger*, 1938] computed for a geomagnetic field intensity (T) of 0.59 oe. Triangles in 4*a* indicate aphanitic, granophyre-bearing border phase.

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netization versus magnetic susceptibility. This form of representation has been found useful to point out probable differences in age between rocks of a single lithology [*Strangway*, 1964], and it also has been used to indicate lithological differences in rocks of identical age [*Beck*, 1966].

The Beaver River gabbro (Figure 4*a*) shows a large variation in magnetic intensity, which is surprising in view of its relatively modest range of chemical differentiation. It varies markedly in texture, however, and in a rough way magnetization appears to correlate with textural features. Relatively weak magnetization is found in nonophitic, fine-grained specimens from near the lower contact but is replaced by generally stronger magnetic intensities as both distance above the base and average diameter of ophitic 'patches' increase. Marginal facies from the upper part of the unit have strong-to-moderate magnetic intensities. There is no obvious correlation between intensity of magnetization and specific gravity, micropegmatite content, or degree of deuteric alteration. This observation stands in marked contrast to results [*Beck*, 1966] from large differentiated sills of very similar composition studied in southeastern Pennsylvania. Perhaps the volatile fraction of the Beaver River gabbro was selectively concentrated into the residual magma during crystallization, thus lowering viscosity and at the same time increasing oxygen fugacity. This in turn might be expected to affect both the size of the ophitic pyroxene grains and the gross content of magnetic oxide minerals. If this result does occur, the volatiles must largely have escaped; with the exception of a few pegmatitic patches, they have left little evidence of their presence in the rock.

The Beaver Bay ferrogabbro varies markedly in chemical composition but very little in grain size or texture. It also shows much less internal variation in magnetic intensity (Figure 4*b*) than do either of the other gabbro units. The sense of variation is also somewhat surprising; both intensity of natural remanent magnetization and magnetic susceptibility decrease as the total iron content of the rock increases. This seemingly anomalous result can be understood in terms of *Osborn*'s [1959, 1962] explanation for the development of iron-rich rocks from ordinary basaltic magma [see also *Carmichael and Nicholls*, 1967]. According to Osborn, ferrogabbros are the result of magmatic differentiation under conditions of constant composition, which requires a progressive decrease in oxygen fugacity. Thus, even though total iron increases during crystallization, the amount of ferric iron available to form magnetic oxide minerals might actually decrease. *Gehman* [1957, Table 7] has determined modes for the Beaver Bay ferrogabbro that show an over-all decrease in the opaque fraction of the rock as total iron increases, which tends to support this idea.

The Black Bay gabbro (Figure 4*c*) shows dominantly high magnetizations with a great deal of scatter, as perhaps should be expected in a rock unit with chemical affinities to gabbro pegmatite. Some of the scatter may be due to the large grain size of the Black Bay gabbro in relationship to the volume of a standard paleomagnetic specimen. To test this idea we effectively doubled the size of the specimens by averaging measurements on two cores from each sample. This procedure resulted in a small decrease in scatter, without affecting the overall high magnetization of the unit.

The evidence available to date clearly indicates that strong magnetization contrasts exist within the Beaver Bay complex and its various subdivisions and that these contrasts are linked to the processes responsible for igneous differentiation, although the nature of the link is not obvious in all cases. In interpreting magnetic surveys involving the Beaver Bay complex [see *Beck et al.*, 1968, for another example], the assumption of 'uniform magnetization' implicit in most methods of depth estimation appears likely to lead to appreciable inaccuracy.

Paleomagnetism

Because of possible effects of magnetic anisotropy, samples from the Beaver Bay ferrogabbro were not used to calculate the average direction of magnetization for the Beaver Bay complex. The remaining twenty-nine sites located in Beaver River and Black Bay gabbros give the average direction of magnetization shown in the last line of Table 1. The paleomagnetic pole position for the Beaver Bay complex was calculated from this orientation of the geomagnetic field by the usual methods and is given at the bottom of Table 2, where it can be compared with pole

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positions for Precambrian igneous rock units exposed elsewhere in the Lake Superior area. Confidence limits (95% probability) for the Beaver Bay complex paleomagnetic pole are $\delta p = 3.5°$, $\delta m = 5.5°$.

The appearance of streaking shown by the Black Bay gabbro in Figure 2*a* requires some comment. This illustration suggests the possibility that components of viscous magnetization in the direction of the present geomagnetic field have been acquired by some Black Bay gabbro specimens and are still present to some degree despite magnetic cleaning. We tend to discount this possibility and have included the Black Bay gabbro in our pole-position calculations, for the following reasons:

1. Streaking should manifest itself in within-site scatter as well as in between-site scatter, especially in sites showing strongly diverse lithology. Two sites in the Black Bay gabbro were selected specifically because they offered exceptional local variations in lithology. Five samples were collected from each of these sites. As expected, samples from these sites exhibited large differences in intensity of remanent magnetization, magnetic susceptibility, and response to alternating-field magnetic cleaning. NRM directions at both sites were widely but randomly scattered, with no evidence of streaking. Magnetic cleaning reduced the greater part of this scatter. The remaining sites in the Black Bay gabbro consist of too few samples to test for streaking.

2. The appearance of streaking is introduced mainly by two sites with high inclinations, which fall within a few tens of degrees of the present direction of the geomagnetic dipole field. Both sites have been magnetically cleaned in peak fields of up to 300 oe, and each has shown moderate magnetic stability, with no tendency to move away from the direction of the present geomagnetic field. Any viscous magnetization present at these sites, therefore, must have an improbably high coercivity. Further information on the magnetic stability of Beaver Bay complex rocks will be provided by mineralogical studies now in progress. Preliminary results of X-ray studies indicate that the composition of magnetic minerals in Black Bay gabbro samples is indistinguishable from the composition of magnetic minerals from other Beaver Bay intrusive rocks.

As shown in Figure 5, the paleomagnetic pole positions for igneous rock units with negative (normal) magnetic polarity form a tight cluster centered at 30°N, 172°W. In contrast, the few published pole positions for nearby igneous rock units of presumed Keweenawan age that have positive (reversed) polarity are scattered over a wide range of latitude and longitude in the area east and north of the cluster of negative poles. Recent unpublished data, summarized below, emphasize this distinction. Preliminary results from intrusive rocks in northern Minnesota, described by most investigators as part of the Duluth gabbro complex, show positive polarity with a paleomagnetic pole located about 20° north of the cluster of negative poles (Table 3). Equally significant are two pole positions computed by K. G. Books (written communication, 1968) for positively magnetized lava flows at the base of the Keweenawan volcanic sequence in northern Minnesota (P of Figure 4) and near Ironwood, Michigan (I of Figure 4). These pole positions also fall north and east of the cluster of negative poles.

The asymmetry in direction of magnetization between units of opposite magnetic polarity apparent in Figure 5 was noted recently by *Palmer* [1968], who attributed it to the effects of

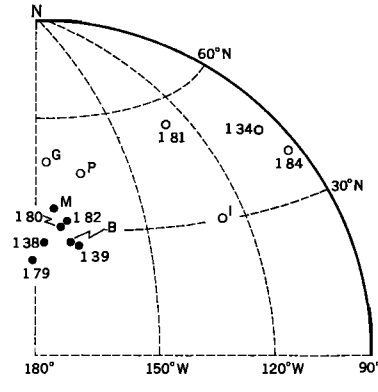


Fig. 5. Paleomagnetic pole positions for igneous rock units from the Lake Superior area. Open circles indicate positive (reverse) polarity. Symbols explained in Table 2, with the exception of I (flows near Ironwood, Michigan) and P (flows near Grand Portage, Minnesota), which are from K. G. Books (written communication, 1968). For pole G, see Table 3. The Portage Lake lava series pole calculated by *Vincenz* [1968] is given in Table 2 but is omitted from Figure 5 because it duplicates pole 1.39.

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TABLE 3. Directions of Magnetization of Positively Magnetized Gabbro, Cook County, Northern Minnesota

Station	Sampling Site λ, °N	Sampling Site φ, °W	N	D, deg	I, deg	k	α_{95}
K104	48.02	90.37	5	136.0	−54.5	104.0	7.5
K106	48.05	90.52	5	145.5	−67.0	11.0	24.0
K107	48.05	90.49	6	107.0	−50.5	36.5	11.0
K108	48.04	90.47	6	118.0	−42.5	78.5	7.5
K109	48.04	90.45	6	117.0	−54.0	120.4	6.0
K110	48.03	90.43	5	121.5	−49.0	95.2	8.0
K111	48.03	90.42	6	147.0	−63.0	67.0	8.5
			Mean of Site-Means				
	48.04	90.45	7	125.0	−55.0	47.0	7.5

Explanation

λ and φ are latitude and longitude, respectively. N is the number of specimens per site (total number of sites in the site-mean calculation). D, I, k, and α_{95} are explained in Table 1. All specimens were cleaned in alternating magnetic fields of 75-oe peak intensity.

Paleomagnetic pole position (G of Figure 5): λ = 48°N; φ = 177°W.

secondary magnetization. We tend to discount this possibility and to prefer explanations involving differences in age, for the following reasons:

1. Within both positively and negatively magnetized groups are units ranging from aphanitic extrusive rocks to very coarse-grained, highly differentiated gabbros. It seems likely that rock units of such contrasting lithology and cooling history would also show sharp contrasts in their vulnerability to secondary magnetization. Thus, if secondary magnetization were responsible for the asymmetry, we would expect the negatively magnetized rock units to be scattered in much the same fashion as are the positive units in Figure 5.

2. If secondary magnetization were important, we would expect to see within-site streaking under favorable conditions. For instance, the sites shown in Table 3 contain as many as six samples, and these samples vary greatly in lithology and grain size in all but a few cases. None of these sites shows noticeable streaking, either before or after af demagnetization.

There is some field evidence that rock units with negative magnetic polarity as a whole are younger than the positively magnetized rock units, but this is by no means certain. Two lines of evidence bear on this subject. (1) Books' suites of positively magnetized lava flows underlie flows of negative polarity, such as the Portage Lake lava series (pole 1.39) and the basalts in the Copper Harbor conglomerate (pole 1.38). Negative polarities extend to the top of the Keweenawan volcanic sequence, whereas lava flows with positive polarity seem to be confined to its lower portions. Negatively magnetized intrusive rocks within the flow sequence (poles 1.79, M and B of Figure 4) also must be younger than the basal part of the volcanic sequence. (2) The Logan diabase intrusions (poles 1.81 and 1.82) were subdivided by *DuBois* [1962, p. 45] entirely on the basis of magnetic polarity. He presents evidence that indicates, however, that the sills comprising most of his group I (positive polarity) are cut by dikes included in his group II (negative polarity). As the pole position for the group II diabase bodies is nearly identical to the poles for all other negatively magnetized igneous rock units in the Lake Superior area (indicating that they were magnetized in roughly the same orientation of the geomagnetic field), it seems reasonable to conclude that the diabase of DuBois' group I is older than any of the igneous rock units represented in the cluster of negative poles.

The age of the Baraga County, Michigan, dikes (pole 1.34) relative to the negatively

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magnetized rock units is not well established. The dikes intrude metamorphosed Huronian sedimentary rocks and in turn are truncated by sedimentary rocks of Cambrian age [*Graham*, 1953, p. 244], pointing to a late Precambrian age. *DuBois* [1962, p. 59] correlated the Baraga County dikes with the Logan diabase, group I, and with the Alona Bay lavas, on the basis of a similarity in paleomagnetic pole position. The relative age of the positively magnetized gabbroic rocks in northern Minnesota (pole G and Table 3) is also not well known. These rocks are present near the base of the Keweenawan sequence and may have intruded the Rove slate (pre-Keweenawan). A potassium-argon age determination of 1060 m.y. was reported by *Goldich et al.* [1961, p. 54, 176] for a sample of Rove slate metamorphosed by a diabase sill, but paleomagnetic samples from what we believe to be this sill have a negative polarity.

Despite some uncertainty about the relative age of a few units, it appears that the igneous rock units of supposed Keweenawan age in the Lake Superior area may fall into two groups: an older group characterized by positive magnetic polarity and scattered paleomagnetic poles and a younger group with negative polarity and tightly grouped poles. At present it is impossible to say whether or not this situation represents a feature of the Keweenawan geomagnetic field (e.g., a period of rapid apparent polar wandering followed by a period of comparative quiescence, similar to the situation described by *Irving* [1966] for the late Paleozoic of Australia). An equally plausible explanation is that the negatively magnetized material accumulated in a very short length of time in comparison with the interval represented by the positively magnetized rock units. The latter possibility might be tested by a program of radiometric age dating coordinated with studies of magnetic polarity.

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